DATA MINING INTRODUCTION

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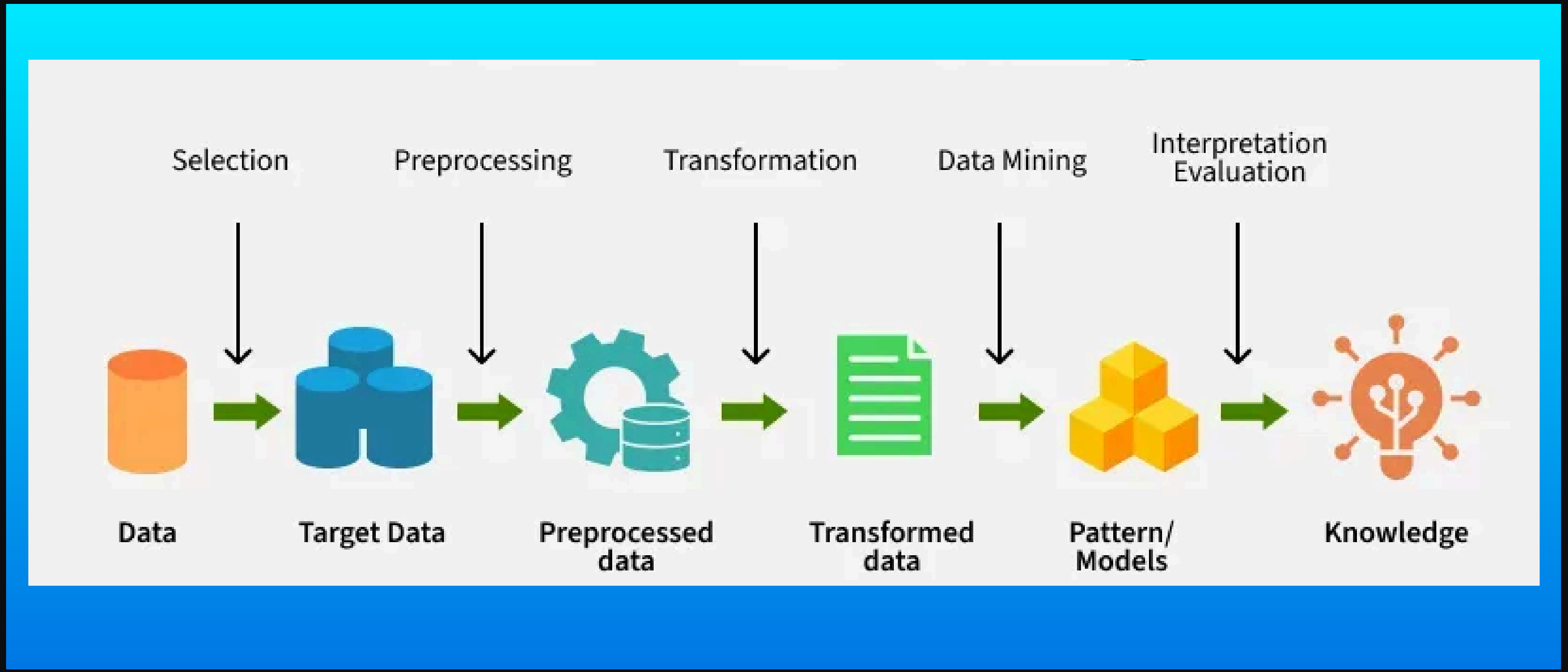
WHAT IS DATA MINING



In today's digital age, data is a critical asset. Effective data analysis enables organizations to identify trends, improve operations, and discover new opportunities. It bridges the gap between data collection and strategic implementation.

- Data mining is a part of Knowledge Discovery in Databases (KDD)
- It works on large datasets stored in databases, warehouses, or cloud storage
- Main goal is to find patterns that are not easily visible.

Real-Life Examples

- E-commerce (Amazon/Flipkart): Product recommendations
- Banking: Fraud detection and credit risk analysis

TYPES OF DATA

Based on Structure

(A) Structured Data:

Data stored in a fixed format (rows & columns).

Examples:

- Student records (Roll No, Name, Marks)
- Bank transaction table

(b)Semi-Structured Data:

Data does not follow a strict table format but still has some structure.

Examples:

- XML files
- JSON data

C) Unstructured Data

Data that has no fixed structure.

Examples:

- Images
- Videos

Based on Data Source

(A) Database Data

Stored in relational databases.

Example: Tables in MySQL

B) Transactional Data

Data generated from daily activities.

Example: ATM transactions

(C) Data Warehouse Data

Large historical data stored for analysis and reporting.

Example: Company sales reports over 10 years

(D) Web Data

Data from websites and social media.

Example:

- Website clicks

Based on Attribute

(A) Nominal Data

Categorical data without order.

Example::-Gender (Male/Female)

(B) Ordinal Data

Categorical data with order or ranking

.

Example::-Student Grades (A, B, C)

(C) Interval Data

Numerical data where difference matters but no true zero.

Example::-Temperature in Celsius (0°C is not absence of temperature)

D) Ratio Data

Numerical data with a true zero.

Example::-Height, Weight

MAJOR DATA MINING FUNCTIONALITIES

1. Classification

Classification is a supervised learning technique where data is categorized into predefined classes.

Example
- Email → Spam / Not Spam
- Student → Pass / Fail

✓ Output is a class label.

2. Regression

Regression is used to predict a continuous numerical value.

Example
- Predict house price
- Predict salary

✓ Output is a number, not a category.

3. Clustering

Clustering groups similar objects into clusters without predefined labels.
(It is an unsupervised technique.)
Example
- Group customers based on shopping behavior

✓ Output is a group/cluster.

4. Association Rule Mining

Association rule mining finds relationships between items that occur together frequently.

Example
In shopping:
- Bread → Butter

✓ Used in Market Basket Analysis.

5. Outlier / Anomaly Detection

Outlier detection identifies data points that are very different from the normal pattern.
Example
- Fraud transaction in bank
- Unusual login attempt

✓ Helps in security and risk analysis.

6.Prediction

Prediction means forecasting future outcomes based on past data.
Example
- Predict stock price
- Predict rainfall

✓ It can be done using classification or regression models.

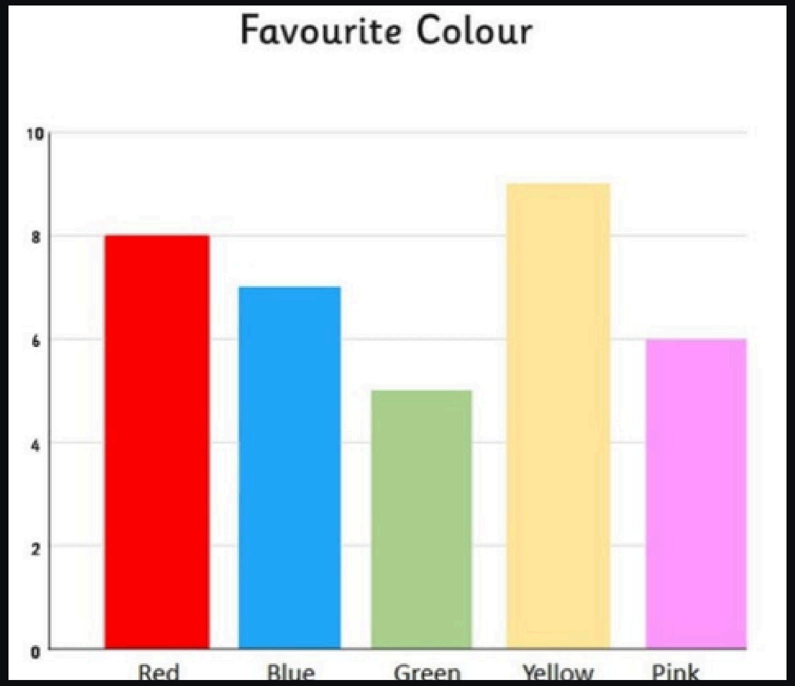
DATA VISUALIZATION

- Data Visualization is the graphical representation of data using charts, graphs, and plots so that users can easily understand information and patterns.

- **Common Types of Data Visualization**

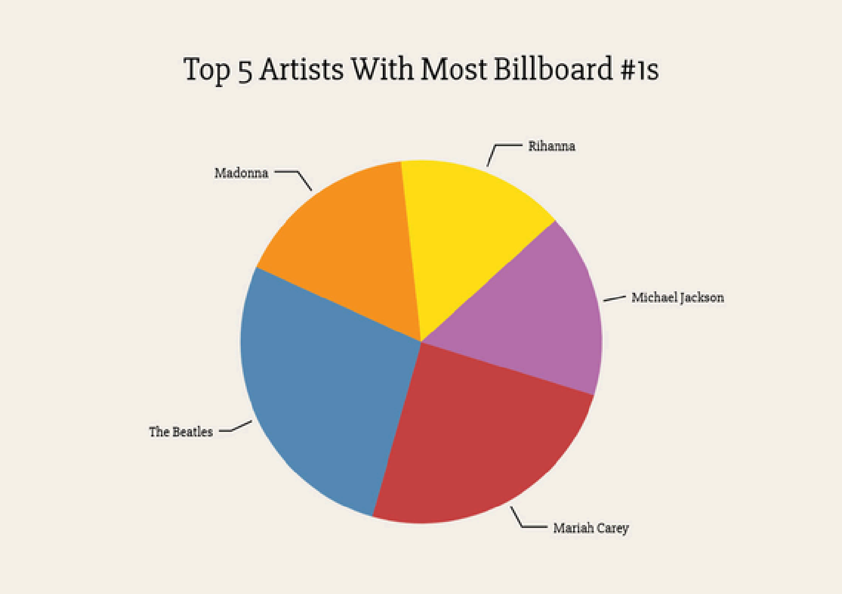
1. Bar Chart

Used for comparing categories.
Example:-Sales of different products.



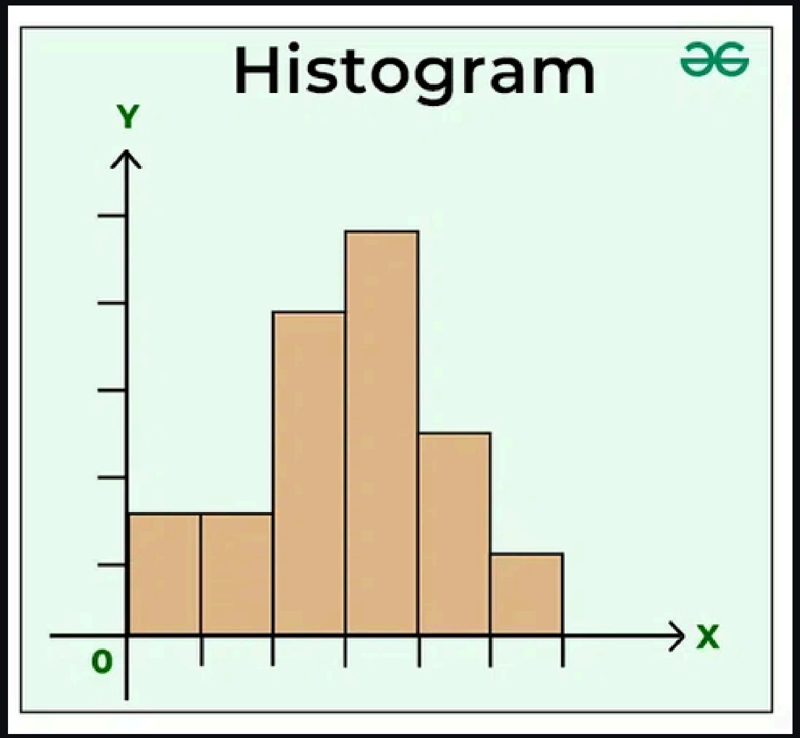
2. Pie Chart

Used to show percentage distribution.
Example:-Market share of companies.



3. Histogram

Used to show frequency distribution of numerical data.
Example:
Marks distribution of students.



DATA PREPROCESSING

Data Preprocessing is the process of converting raw, incomplete, and noisy data into a clean and usable format for data mining and analysis.

1. Data Cleaning

Removes errors and improves quality of data.

- Handling missing values
- Removing duplicates
- Noise removal (incorrect data)
- Correcting inconsistent values

2. Data Integration

Combines data from multiple sources into one dataset.

- Merge databases/files
- Remove redundant data
- Resolve data conflicts

3. Data Transformation

Converts data into a suitable format for mining.

- Normalization (scaling values)
- Aggregation (summarizing data)
- Encoding categorical values

4. Data Discretization

Converts continuous data into categories/ranges.

Example:

Age → (0–18), (19–35), (36–60)

5. Data Reduction

Reduces size of dataset while keeping important information.

- Dimensionality reduction
- Attribute subset selection (feature selection)
- Compression and sampling

ATTRIBUTE SUBSET SELECTION, SAMPLING & CLUSTERING

1. Attribute Subset Selection (Feature Selection)

Attribute Subset Selection is the process of selecting the most important attributes (features) from a dataset and removing unnecessary or irrelevant ones.

Example:
For predicting student performance, useful attributes may be:

- attendance
- internal marks
- assignment score

Unnecessary attributes may be:

- student ID
- phone number

2.Sampling

Sampling is the process of selecting a small representative subset of data from a large dataset for analysis.

Example:
Instead of analyzing 10 lakh customer records, we analyze a sample of 10,000 records.

Types of Sampling

Random Sampling

- Data selected randomly
- Each record has equal chance

Stratified Sampling

- Data divided into groups (strata)
- Samples taken from each group

3. Clustering

Clustering is an unsupervised technique used to group similar data points into clusters.

Important:
No predefined labels/classes are given.

Example of Clustering
A company divides customers into:

- Cluster 1: High Income, High Spending
- Cluster 2: Medium Income, Medium Spending
- Cluster 3: Low Income, Low Spending

This helps in better marketing strategies.

THANK YOU

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